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Palaeomagnetism of the Early to Mid-Ordovician Salala igneous ring complex, Red Sea Hills, Sudan

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SUMMARY

Palaeomagnetic samples were collected from the Early to Mid-Ordovician igneous ring complex at Salala, northern Red Sea Hills, Sudan. Detailed demagnetization experiments on 114 samples from 17 sites yield two distinct groups of characteristic directions of the natural remanent magnetization. Component 'A', predominant in samples from the quartz syenitic central part and minor ring dyke of the complex, displays rather high blocking temperatures, well above 600 °C and is presumably carried by haematite. The direction of this component (Dec/Inc = 1°/+37°) is virtually identical with the direction of the present day Earth's magnetic field at the sampling locality (Dec/Inc = 0°/+37°). Mixed polarities of 'A' magnetization strongly suggest a Recent/pre-Brunhes age of this component, and the effects of long-term weathering and low-temperature oxidation. Component 'B', pointing to the southeast and upward, has been identified in seven sites from the major ring dyke, composed of aegirine syenite. Maximum blocking temperatures of this component do not exceed 580 °C and, together with the results of rock magnetic and optical examinations, are indicative of primary magnetite as the carrier of the characteristic magnetization. The mean direction (Dec/Inc = 124°/−50°, $\alpha_{95} = 9.2$, $k = 44.1$) differs significantly from component 'A' and is thought to be primary and representative for the direction of the Earth's magnetic field during emplacement and cooling of the ring complex in Mid–Ordovician times. The resulting palaeomagnetic pole (40°N, 30°W, $dp = 8$, $dm = 12$), situated off the coast of West Africa, plots very close to a cluster of Ordovician palaeomagnetic poles from Africa. These poles have not been integrated into the majority of the various apparent polar wander paths for Gondwana but this new result re-enforces their correctness. The application of recently revised rotation parameters for eastern Gondwana significantly reduces the longitudinal gap between the African data and that from eastern Gondwana (particularly Australia) for the Ordovician.

Key words: Africa, apparent polar wander path, Palaeozoic, palaeomagnetism, ring complexes.

1 INTRODUCTION

It is now widely accepted that the position of the palaeo southpole has shifted from a position close to the occurrence of Late Ordovician tillites in the northern Sahara to a Permo–Carboniferous position off Southern Africa, corresponding to the widespread occurrence of Late Palaeozoic diamictites in the lower parts of the Dwyka sequence of the Karoo basin. This conclusion is based on both the distribution and age of glacial deposits on the Gondwana continents (see Caputo & Crowell 1985 and references therein) and on palaeomagnetic results (Briden 1968; McElhinny & Opdyke 1968; Kent & May 1987, Van der Voo 1988). However, this rather loosely defined framework allows for several conflicting solutions for the motion of Gondwana during Early and Mid–Palaeozoic times depending on age, and structural interpretations and degree of reliability assigned to individual palaeomagnetic studies. The alternative solutions for the Palaeozoic APWP for Gondwana (Fig. 1) have far reaching implications for its Ordovician palaeogeography.

Although approximately 50 individual palaeomagnetic

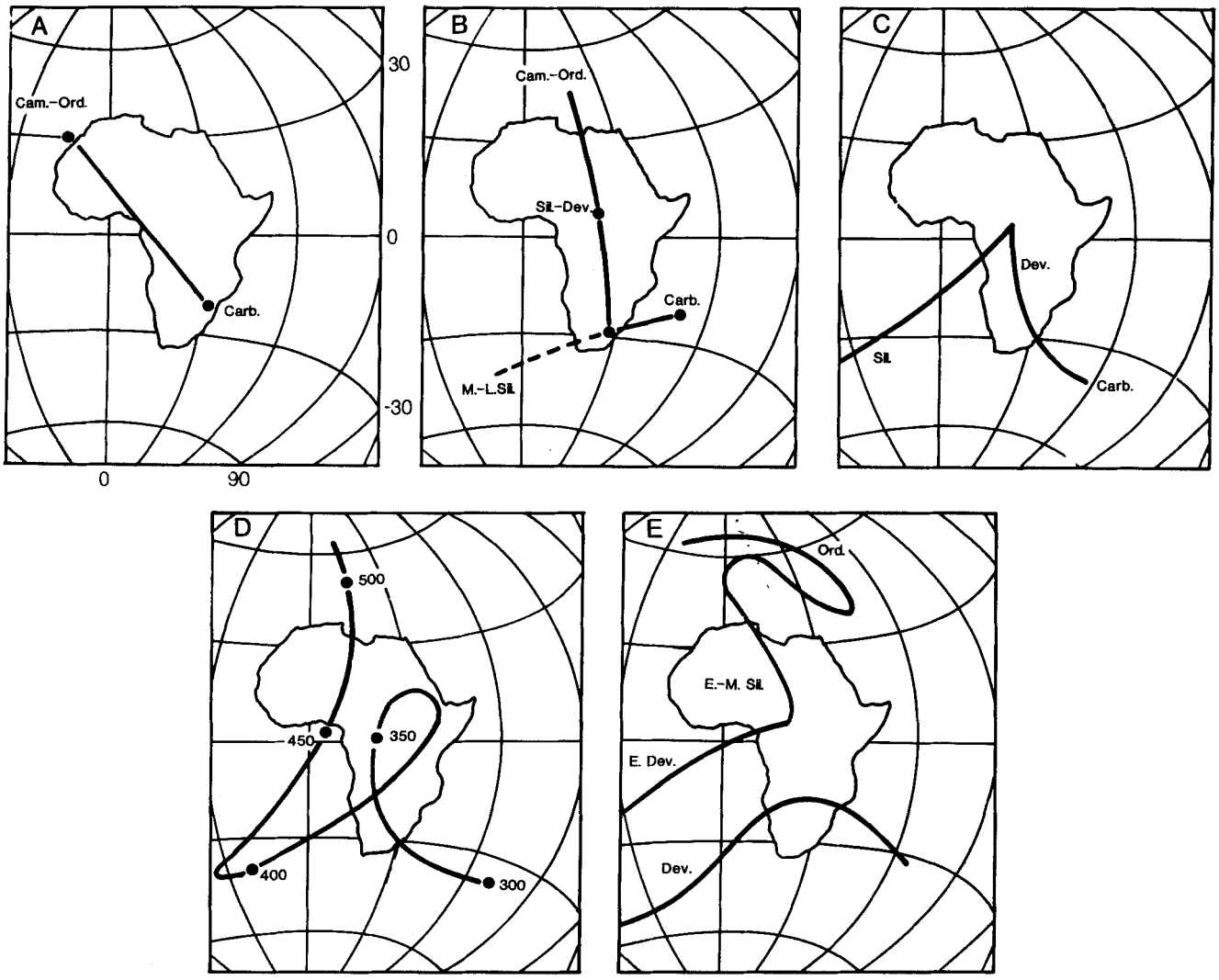


Figure 1. Proposed apparent polar wander paths for Gondwana. (a) according to McElhinny *et al.*, (1968), (b) McElhinny & Embleton (1974), (c) Schmidt & Morris (1977), (d) Morel & Irving's (1978) path 'Y', (e) Goleby (1980). Note that path, (d) and (e) rely almost exclusively on Eastern Australian palaeomagnetic poles, while path 'B' represents the combination of Australian and African data in a conventional Smith & Hallam (1970) reconstruction. (b)–(e) After Livermore, Smith & Briden (1985).

pole positions have been published for the period from Cambrian to the Devonian for Africa, South America, Australia and Antarctica (e.g. review by Hargraves, Dawson & Van Houten 1987), only a small minority of the data actually withstand the scrutiny of modern reliability tests such as those devised by McElhinny & Embleton (1976) and, more recently, by Van der Voo (1988). Reasons for the apparently poor quality of the data set are manifold. Continental red beds which have been some of the prime targets of palaeomagnetic studies on the Gondwana continents usually lack distinct fossil assemblages and are consequently poorly dated. A striking example is the Australian Mereenie sandstone (Embleton 1981), where a very well determined palaeomagnetic result is limited in its usefulness by the extremely wide possible stratigraphic age range of the rock unit (Ordovician to Devonian). Similar problems are also known to be associated with the majority of the South American palaeomagnetic data as compiled by Vilas (1981). In the absence of stability tests, such as the fold test, dating the acquisition of the natural remanent magnetization (NRM) is difficult. The rather complex multicomponent character of the magnetization of some previously studied Lower Palaeozoic rocks from Africa [such as the Ordovician part of the South African Table Mountain Group (Graham & Hales 1961)] has been identified only recently (Bachtadse, Van der Voo & Haelbich 1987) and it can be assumed that data on terrestrial sediments from Mauritania (Morris & Carmichael 1978; Kent, Dia & Sougy 1984; Perrin, Elston & Moussine-Pouchkine 1988) do indeed reflect Mid- to Late Palaeozoic overprints rather than primary directions of the NRM (Kent & May 1987). Further difficulties arise from the open question whether Palaeozoic palaeomagnetic data from eastern Australia should be regarded as being representative for eastern Gondwana (Schmidt & Morris 1977; Goleby 1980) or rather reflect the palaeomagnetic history of displaced terranes, now incorporated into the east Australian Palaeozoic Lachlan Fold Belt (McElhinny &

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Embleton 1976). Because of the uncertainties surrounding many of the data only the pole position for the Early Ordovician Ntonya Ring Complex, Malawi (Briden 1968) and to a lesser extent the results from Damaran granites, Namibia (Corner & Henthorn 1978), and the Hook intrusives, Zambia (Brock 1967) have so far been accepted as calibration poles (McElhinny 1973; Kröner *et al.* 1980) for the Lower Palaeozoic of Africa. It is evidently important to strengthen the database.

In this paper we report results of a palaeomagnetic study from the Ordovician Salala Ring Complex in northern Sudan (21.36°N, 36.13°E). The rocks have been dated previously (Rb/Sr, Cavanagh 1979) and their. Early to Mid–Ordovician age has been established. Furthermore, the results of a palaeomagnetic pilot study (Cavanagh 1979) hinted towards the presence of an ancient component of the magnetization in some samples, which did not resemble any African post-Ordovician direction, therefore suggesting the palaeomagnetic potential of the Salala Ring Complex.

2 GEOLOGICAL SETTING AND SAMPLING TECHNIQUES

Anorogenic, intraplate igneous ring complex provinces are a prolific feature of African Phanerozoic geology. Although they occur preferentially in areas deformed by the latest Precambrian Pan–African orogeny they are unrelated to plate margin activities (Bowden *et al.* 1987 and older references therein). Individual high-level intrusive ring complexes range from isolated plug-like intrusions (a few hundred meters in diameter) to large heterogeneous circular structures several tens of km across (Bowden 1985). Although the age of those igneous complexes covers the whole Phanerozoic, they are episodically concentrated into four periods (Vail 1985): Mid–Ordovician to Early Silurian, Mid–Triassic, Mid–Jurassic to Mid–Cretaceous and Early Cenozoic. In the area of the Sudanese Red Sea Hills (Fig. 2) at least 100 ring complexes of Early Palaeozoic and Mesozoic age have been described (Vail & Kuron 1978; Vail 1983).

Being relatively accessible, the ring structure of Salala, about 200 km north of Port Sudan (Fig. 2), has been mapped in detail (Jar-en-Nabi 1976), and is the subject of a comprehensive geochronological as well as pilot palaeomagnetic study (Cavanagh 1979).

The ring complex itself is comprised of a ring doublet (Fig. 2). The major ring dyke, about 8 km wide, is composed of aegirine syenite, while the minor ring dyke, about 4 km in diameter, is built up by quartz syenite. The minor ring dyke encloses an anorthositic gabbro-diorite, which in turn is intruded by a central nepheline syenite plug.

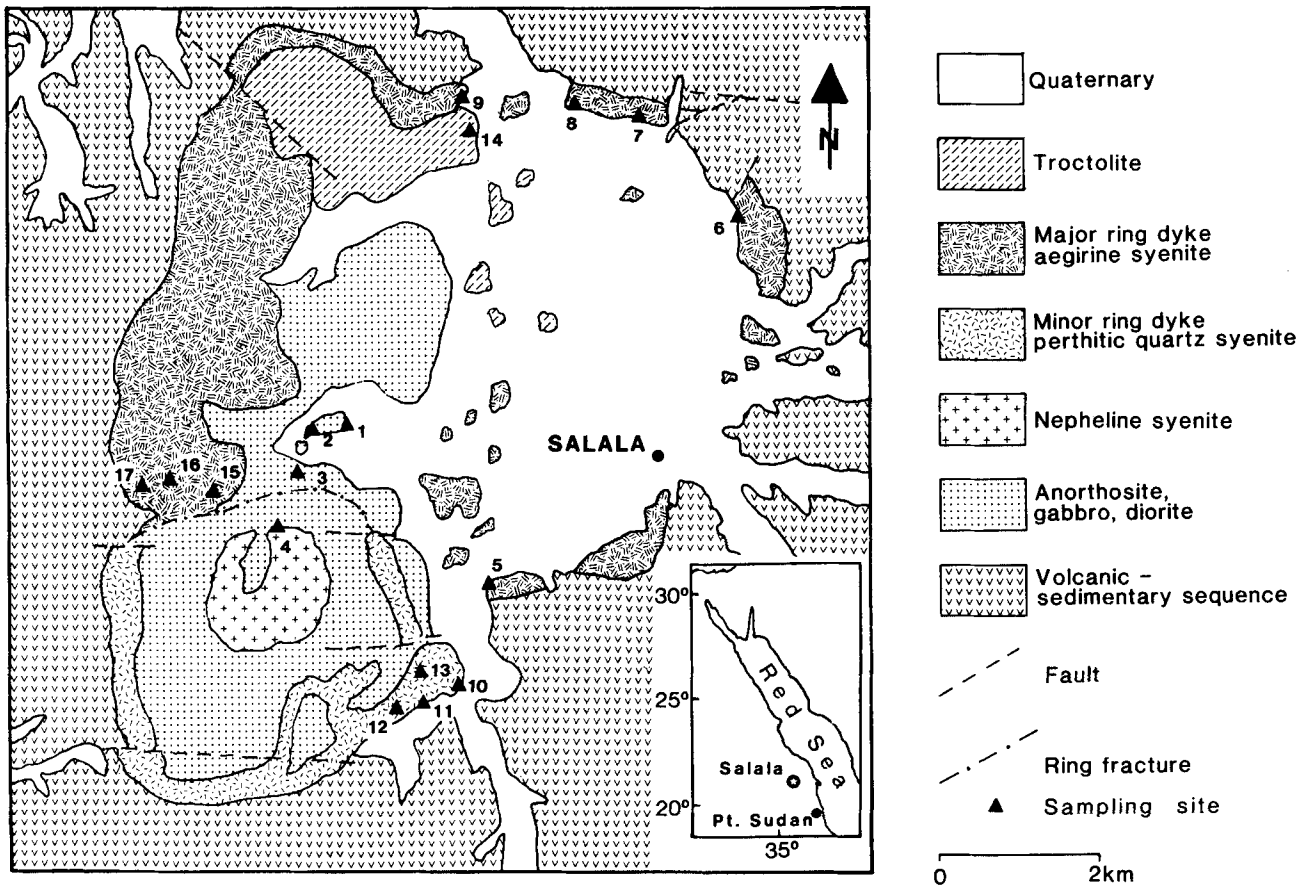


Figure 2. Geological sketch map of the Salala Ring Complex. Modified after Jar-en-Nabi (1976).

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Several occurrences of troctolite are also enclosed by the major ring dyke. Geochronological studies (Cavanagh 1979) yielded Rb/Sr isochron ages of 450 ± 16 Ma for the major ring dyke, 469 ± 9 Ma for the minor ring dyke and 481 ± 11 Ma for the central gabbro-diorite (Rb^{87} decay constant: $1.42 \times 10^{-11}\,yr^{-1}$). Field evidence indicates that the minor ring dyke postdates the major ring dyke, and that both postdate the central gabbro-diorite (Jar-en-Nabi 1976). The combination of the isotopic data and field evidence indicates little or no time lag between the emplacement of the two ring structures; this brackets the age of the major ring dyke between around 460 and 466 Ma to satisfy the 2σ confidence limits of each age determination (Cavanagh 1979).

The various lithologies of the Salala Igneous Ring Complex have been palaeomagnetically sampled at 17 sites, covering the anorthosites, the minor syenite plug, the gabbro-diorite, the minor as well as the major ring dyke and the troctolites (Fig. 2). A total of 114 samples was drilled in the field with a petrol powered portable drill and oriented with a standard sun compass.

3 LABORATORY TECHNIQUES

The oriented core samples, 25 mm diameter, were cut to standard height (23 mm) palaeomagnetic specimens. Directions and intensities of the NRM were measured using a two-axis CCL cryogenic magnetometer surrounded by a μ-metal shield and a set of Helmholtz coils and interfaced to a microcomputer. Thermal and alternating field (AF) demagnetization experiments have been carried out using magnetically shielded equipment. The results of the stepwise demagnetization experiments have been plotted on orthogonal (Zijderveld 1967) or stereographic diagrams, and subsequently subjected to a line and plane fitting procedure (LINEFIND, Kent, Briden & Mardia 1983). Only linear elements based on a minimum of four consecutive measurements and with an α_{95} of less than 20° were accepted as being reliable estimates for the characteristic direction of the NRM (ChRM) or any component of magnetization removed during demagnetization. Characteristic directions from samples from the major dyke usually yielded values for α_{95} smaller than 5°. The resulting site mean directions with α_{95} smaller than 20° are listed in Table 1. In addition, polished thin sections from selected specimens have been studied under reflected and transmitted light in order to determine their petrological character and the carriers of the magnetization.

Table 1. Site mean directions of the ChRM for the Salala Ring Complex, Red Sea Hills (21.36°N, 36.13°E).

Site	Lithology	N/n	Dec/Inc	α_{95}	k	Group
SA1	Anorthosite	10/10	350.6/17.9	9.5	26.6	A
SA2*	Anorthosite	6/6	75.7/46.3	19.2	13.2	—
SA3	Anorthosite	8/8	3.4/35.6	5.6	98.8	A
SA4	Neph. syenite	9/8	1.2/31.9	7.2	60.0	A
SA5	Aeg. syenite	6/6	150.3/−67.2	12.7	28.9	B
SA6	Aeg. syenite	9/8	116.4/−44.7	10.5	29.0	B
SA7	Aeg. syenite	7/7	116.2/−47.5	7.1	72.8	B
SA8	Aeg. syenite	6/6	118.6/−48.4	9.4	51.6	B
SA9	Aeg. syenite	9/9	130.5/−45.9	7.6	46.9	B
SA10	Q. syenite	5/4	4.2/ 56.5	10.6	76.3	A
SA11	Q. syenite	5/5	no within site consistency			
SA12*	bas. dyke	5/5	24.0/75.3	19.9	15.6	—
SA13	Q. syenite	7/7	9.5/57.9	10.3	26.5	A
SA14	Troctolite	7/7	174.8/−41.6	11.4	28.4	A rev
SA15	Aeg. syenite	3/3	0.8/15.6	19.5	35.9	A
SA16	Aeg. syenite	7/5	319.4/41.8	17.8	18.6	B rev
SA17	Aeg. syenite	5/5	107.3/−52.3	13.0	35.8	B

N/n is the number of samples measured and the number of samples used to compute the site mean direction. Dec, Inc: mean declination, inclination; k: precision parameter; α_{95}: semi angle of the cone of confidence about the mean direction at the 95 per cent probability level (Fisher 1953).
* Not used for calculation of mean direction.

4 PALAEOMAGNETIC RESULTS

Typical intensities of the NRM of the samples measured are in the range $0.5–1.5\,A\,m^{-1}$. Only a few samples have significantly higher or lower intensities. The directions of total NRM cluster either around a northerly and downward or southerly and upward mean direction, which is very close or identical to the axis of the present day field (PDF) at Salala (component 'A'), or display southeasterly and moderately upward pointing directions (component 'B'). During the pilot stage of the study, selected specimens of both directional populations were subjected to detailed thermal and AF demagnetization experiments in order to define their directional and rock magnetic properties.

AF demagnetization of samples having NRMs close to the PDF was not successful in isolating a stable direction of the NRM, due to the dominance of a low-coercivity component and the rather viscous behaviour of the samples even at low peak fields (2.5 mT). AF demagnetization of unweathered samples from the major ring dyke with easterly and upward pointing directions proved more successful (Fig. 3a). After removal of a small viscous component at rather low peak fields (15 mT), univectorial decay of the NRM towards the origin of the orthogonal projection is typical, with maximum coercivity of the carriers of the magnetization not exceeding 75 mT. Almost identical behaviour of the magnetization is displayed during thermal demagnetization clearly demonstrating the single component magnetization of samples from the major ring dyke (Fig. 3b). Maximum blocking temperatures of 580 °C, low coercivities as indicated by the results of isothermal remanent magnetization (IRM) experiments (Fig. 4a) as well as by AF demagnetization unambiguously point towards magnetite as the predominant carrier of the characteristic NRM (ChRM). However, the complexity of the observed magnetic behaviour is clearly a function of the degree of weathering. The presence of a high blocking temperature component (up to 680–700 °C), which is very likely to be carried by haematite and which is indistinguishable from the direction of the present day field at the locality, disguises the direction of the ancient magnetization (Fig. 3c and d) to varying degrees. Thermal demagnetization and subsequent vector analysis allows the two components present in the more weathered material from the major ring to be distinguished. The vectors removed in the 450–580 °C range (e.g. Fig. 3c) are generally

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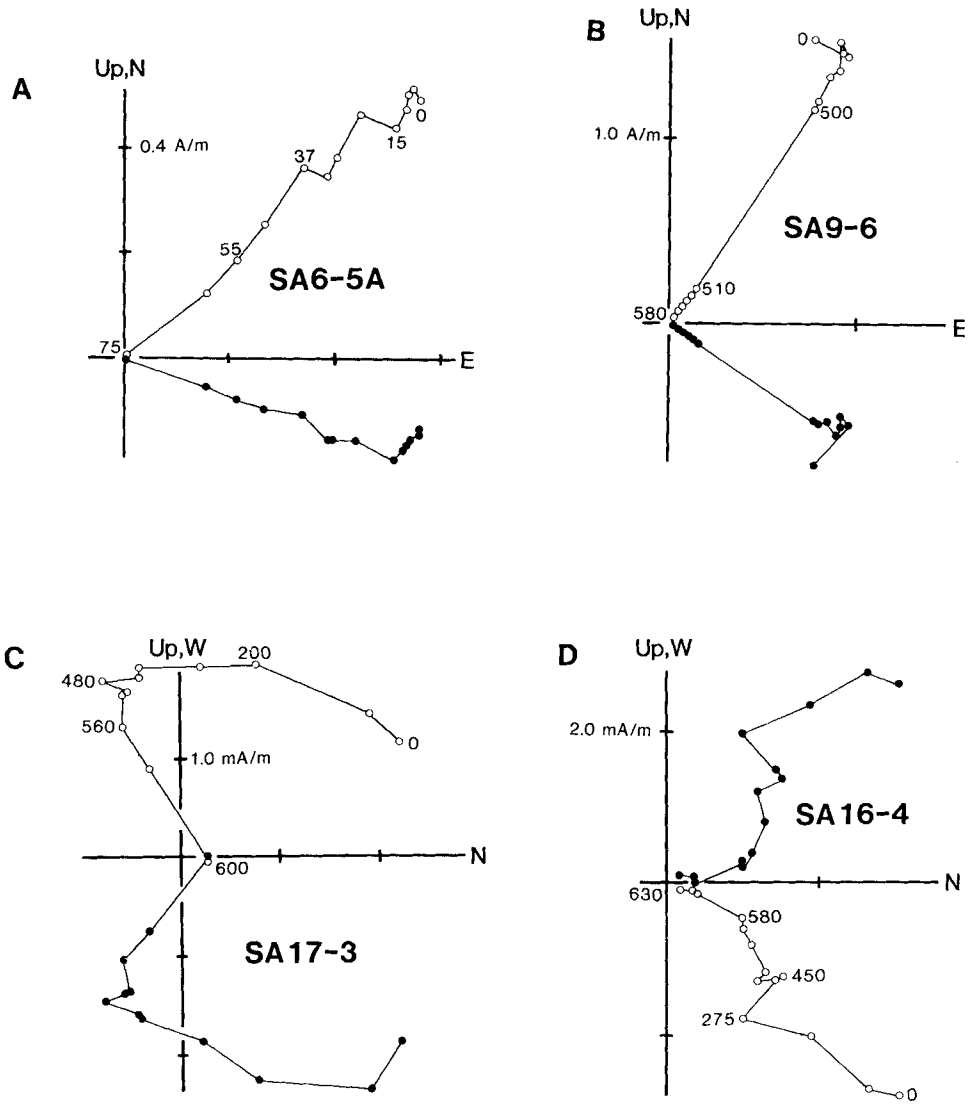


Figure 3. Orthogonal demagnetization diagrams (Zijderveld 1967) for samples from the **major** ring dyke. Closed (open) circles represent projections on the horizontal (vertical) plane. (a) AF demagnetization, (b)–(d) thermal demagnetization. Demagnetization field (mT) and temperature (°C) are marked.

pointing to the east and upward, and are very similar to the characteristic magnetization as identified in unweathered samples in neighbouring rocks. In addition, a characteristic component of magnetization, which is directed anti-parallel to the characteristic directions as described above and has similar blocking temperatures, seems to be present in partially weathered samples from site SA16 (e.g. Fig. 3d). Complete, recent remagnetization has been observed in samples from the minor ring and some other parts of the Ring Complex, where no ancient magnetization could be identified (Fig. 5a and b). The overprint (component 'A') can be of either polarity (cf Figs 5a, b and 7) and is diagnostic of a Quaternary age of remagnetization which may be a result of prolonged weathering.

5 ROCK MAGNETIC AND OPTICAL PROPERTIES

The rock magnetic properties as established during AF and thermal demagnetization and IRM experiments clearly indicate magnetite as the dominant carrier of the ChRM. Substantially higher blocking temperatures as well as higher coercivities as revealed by IRM experiments also suggest the presence of haematite in more weathered rocks from the more central units of the Salala Ring Complex. Optical examination of representative polished thin sections shows the predominant presence of magnetite in various high-temperature exsolution states in all the material studied and only minor amounts of haematite have been observed along the edges or along cracks of individual magnetite grains in samples from the minor part of the ring complex. Additional electron microprobe analysis shows the high-temperature oxidation state of the magnetites. Large ilmenite lamellae (Fig. 6) as predominantly present in samples from the major ring are very likely to be indicative of prolonged cooling. Low-temperature demagnetization at liquid nitrogen temperature in field-free space (Merrill 1970) does not affect the intensity nor the direction of the NRM of samples from the major ring, where the 'B' component has been identified, thus indicating single domain or pseudo

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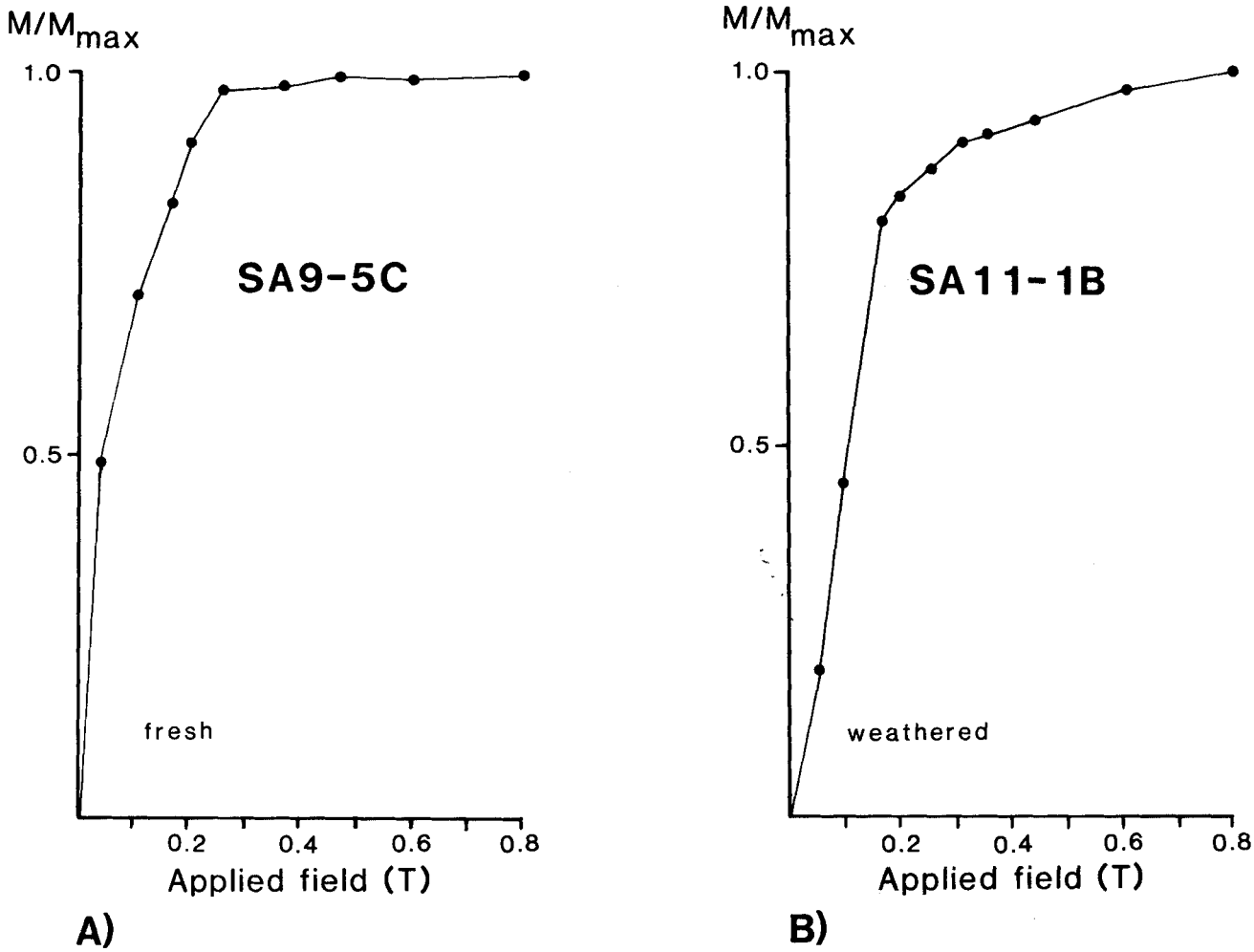


Figure 4. Isothermal remanent magnetization acquistion curves for samples from the **major** (a) and **minor** (b) ring dyke.

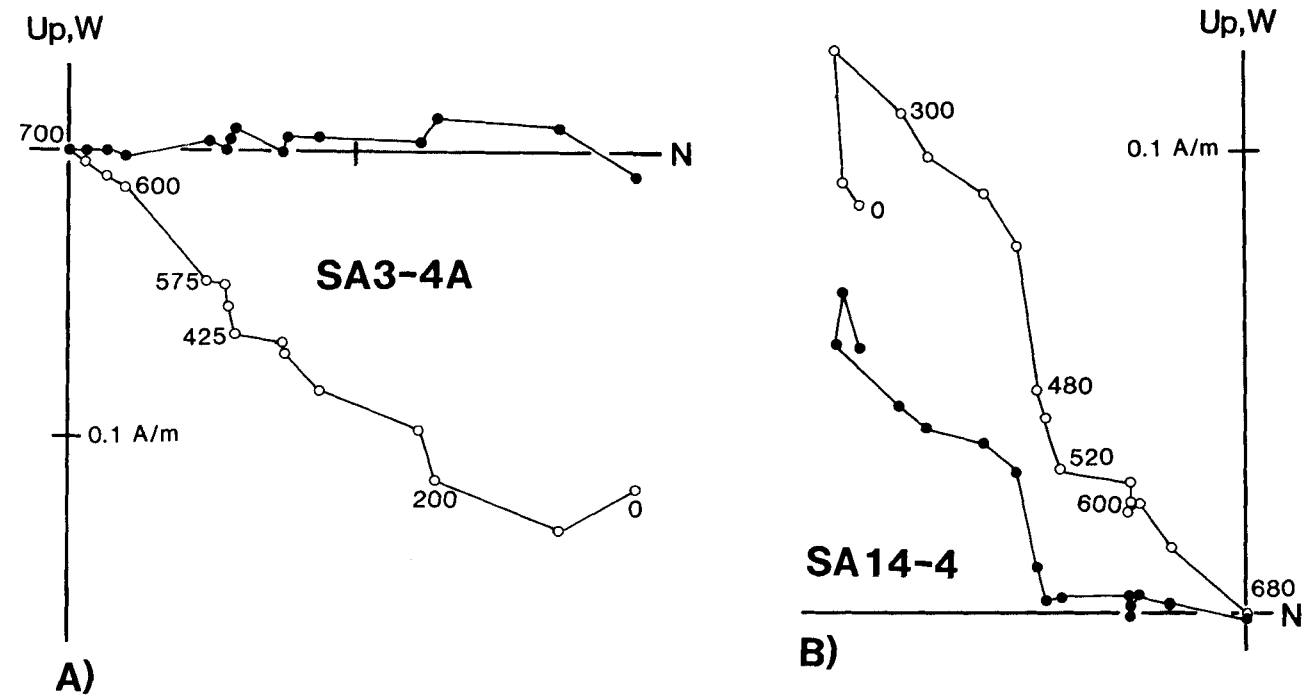
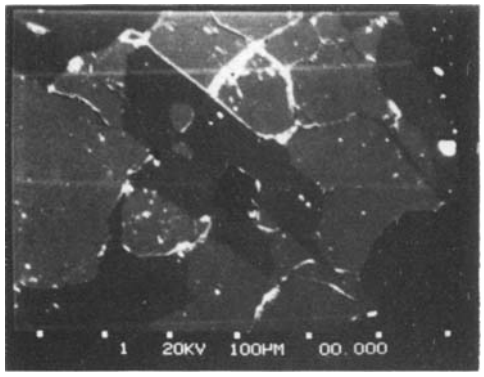


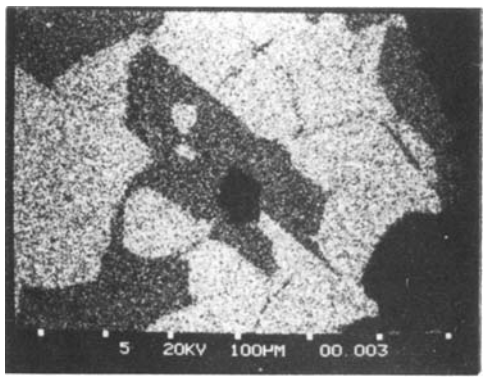
Figure 5. Orthogonal thermal demagnetization diagrams (Zijderveld 1967) for samples from the **minor** ring dyke. Notation as in Fig. 3.

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(a)



(b)



(c)

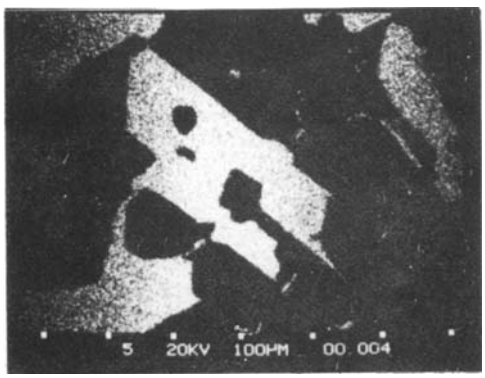


Figure 6. Electron microprobe photomicrograph of a high-temperature oxidized magnetite grain from site SA9 (major ring). Note the dark areas in the central and marginal areas of the grain (a) representing ilmenite. (b)–(c) Back scatter images of the same grain. Light areas in (b) show the predominance of Fe. Light areas in (c) are diagnostic for the presence of Ti. Scale markers at 100 μm intervals.

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single domain state of the NRM-carrying magnetites (Stacey & Banerjee 1974). Samples in which the predominance of the 'A' component has been identified, however, lose up to 50 per cent of the initial intensity of the NRM during cooling, which does not recover during warming in field-free space. This result is diagnostic of the presence of multidomain magnetite. We can therefore conclude that component 'B' is carried by a deuterically exsolved single or pseudo-single domain magnetite, while the PDF overprint (component 'A') as observed in specimens from the minor ring is probably carried by secondary haematite as well as by viscous multi-domain magnetite.

6 CONCLUSION AND DISCUSSION

Characteristic sample directions as well as site mean directions with values of α_{95} smaller than 20° have been plotted on equal area projection (Fig. 7) and listed in Table 1. Pole positions are calculated in Table 2. The close proximity of the majority of the characteristic directions (component 'A') to the normal and reversed direction of the PDF is apparent on both sample and site level (Fig. 7a and b). In addition, a second cluster of characteristic directions pointing towards the southeast and upward, or to the northwest and downward (component 'B') can be identified. The latter is significantly different from the direction of the PDF and has been isolated in a total of seven sites from the major ring dyke (Table 1). The rock magnetic properties as well as the direction of this characteristic magnetization itself argue strongly for an ancient magnetization, interpreted to be representative for the direction of the Earth's magnetic field during rock formation. The occurrence in one site of directions antiparallel to 'B' seems to point towards a prolonged cooling period of the ring complex, at least long enough to record one reversal of the Earth's magnetic field. Consequently, and in accord with our rock magnetic observations, it can be argued that the effects of secular variation on the characteristic direction of the NRM have been effectively cancelled and that the 'B' component is representative of a time averaged geomagnetic field direction. Although mixed polarities do not by themselves prove the primary character of the magnetization, the occurrence of normal and reversed magnetization, supported by the rock magnetic and textural evidence from the rocks from the major ring, strongly suggests a syn-cooling age of the 'B' component.

Table 2. Salala Ring Complex mean directions and palaeomagnetic pole positions.

	N	k	α_{95}	Dec	Inc	Lat (°N)	Long (°E)	dp	dm
Component 'A': Mostly minor ring dyke	7	21.2	13.4	1.1	36.9	88.7	162.9	9.2	15.7
Component 'B': major ring dyke	7	39.6	9.2	124.2	−50.2	39.6	329.5	8.3	12.3

N: number of sites used to compute the mean direction; Dec, Inc: mean declination, inclination; k: precision parameter; α_{95}: semi angle of the cone of confidence about the mean direction at the 95 per cent probability level (Fisher, 1953); Lat, Long: latitude, longitude of the palaeomagnetic pole; dp, dm: semi-minor and semi-major axis of the polar error ellipse at the 95 per cent probability level.

The mean direction for component 'B', based on seven sites from the major ring is 124.2° declination and −50.2° inclination ($k = 39.6$, $\alpha_{95} = 9.2°$). The resulting palaeomagnetic pole position at 39.6°N, 30.5°W is located to the northwest of the west African coast. It is significantly different from the Ordovician 'starting point' of the APW paths of McElhinny & Embleton (1974), Morel & Irving (1978) and Goleby (1980) (Figs 1b, d and e, respectively). It is, however, in good agreement with the 'starting point' of the McElhinny & Opdyke (1968) APW path (Fig. 1a) and other published Ordovician palaeomagnetic pole positions for Africa (Table 3 and Fig. 8). The longitudinal difference

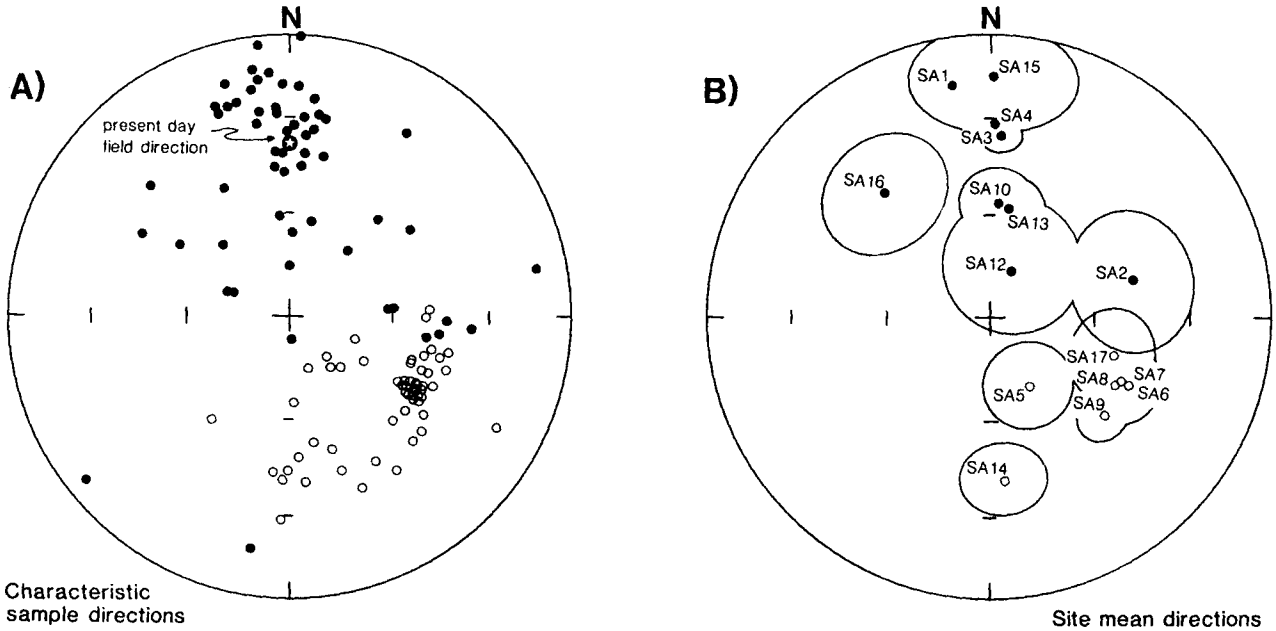


Figure 7. Stereographic projection of characteristic directions with associated α_{95}. (a) Sample means, (b) site means. Open (closed) symbols represent projections on the upper (lower) hemisphere. Labels on (b) refer to individual sites as listed in Table 1.

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Table 3. Ordovician Palaeomagnetic pole positions from Africa, Australia and Antarctica in a Gondwana reassembly.

Formation	Symbol	Age(method)	Lat (°N)	Long (°E)	k	α_{95} (dp, dm)	Reference
Africa							
Damara granites	DG	458 ± 8(Rb/Sr)	27	351		5	Corner & Henthorn (1978)
Hook intrusive	HK	517 ± 18(Rb/Sr)	14	336	13	36	Brock (1967)
Ntonya Ring Complex	NT	485 ± (K/Ar)*	28	345	1045	2	Briden (1968)
Graafwater Formation	GR	Ol(strat.)	28	14	25	9	Bachtadse *et al.* (1987)
Blaubeker Formation	BL	471 ± 26(Rb/Sr)	51	352	—	(10, 19)	Kröner *et al.* (1980)
Salala Ring Complex component B	SA	~460(Rb/Sr)	40	330	—	(8, 12)	this study
Pakhuis/Cedarberg Formations	PAC	Ou/Sl(strat.)	25	343	9	18	Bachtadse *et al.* (1987)
Central and Northern Australia							
Jinduckin Formation	JF	Ol(strat.)	36	359	10	11	Luck (1972)
Stairway Sandstone	SS	Om(strat.)	51	34	25	9	Embleton (1972)
Eastern Antarctica							
Sor Rondane intrusives	SR	~496(Rb/Sr)	11	8	—	5	Zijderveld (1968)
Taylor valley dykes	TV	460 ± 7(Rb/Sr)	36	9	75	11	Manzoni & Nanni (1977)

Palaeomagnetic poles from Australia and Antarctica have been transferred into African coordinates using the rotation parameters given by Lawver & Scotese (1987). Whenever necessary isotopic ages have been recalculated using the appropriate decay constants (Steiger & Jäger 1977). α_{95}: semi angle of the confidence about the mean direction at the 95 per cent probability level (Fisher 1953); Lat, Long: latitude, longitude of the palaeomagnetic pole; dp, dm: semi-minor and semi-major axis of the polar error ellipse at the 95 per cent probability level. O: Ordovician; S: Silurian; l, m, u: lower, middle, upper. * Mean K Ar age; But more recent studies by one of us (J. C. B. & D. C. Rex) suggest the magnetization age of the Ntonya Ring Complex may be rather older.

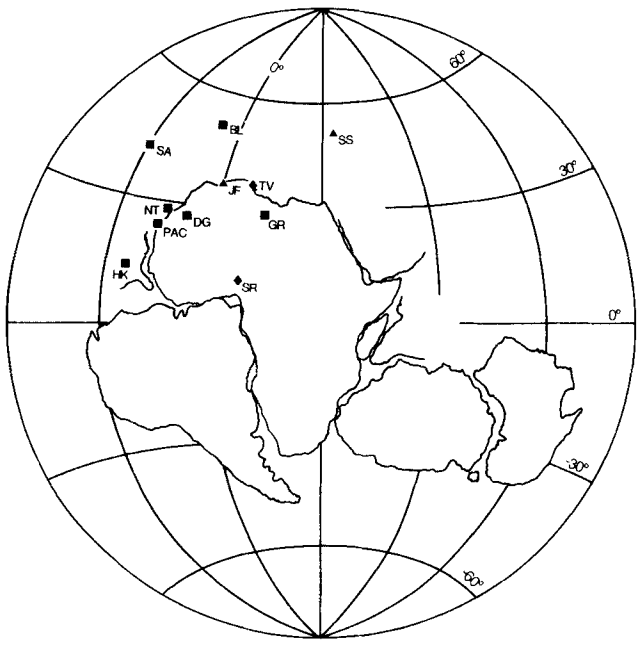


Figure 8. Pole for this study (SA) and other data from Africa (squares), Australia (triangles) and Antarctica (diamonds) listed in Table 2. Gondwana reassembly after Lawver & Scotese (1987).

of ~30° between the African poles and the Ordovician poles from other Gondwana continents in Fig. 1(b, d and e) remains conspicuous. Using finite rotation parameters (Lawver & Scotese 1987) slightly different from the ones used by Smith & Hallam (1970) for the reconstruction of eastern Gondwana results in a tighter fit of Australia, Antarctica and Africa, and reduces the longitudinal gap to approximately 15°. It is not clear whether this discrepancy is significant, and whether it is due to geomagnetic, tectonic or other geological causes.

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